Contacting course staff and other students

Communication between you, the course staff, and other students is important. Please help us keep things efficient by following these guidelines:

- **General questions about course content, assignments, logistics, etc.:** Please use **Inscribe (Q&A Community)** in the left navigation. You can usually expect a response within 24 hours. If your question is specific to you or includes personal information, set the post visibility to **Moderators (instructors) only**. Otherwise, please post to the full class so others can benefit. We encourage you to read and participate in Q&A discussions; we may award extra credit to students who are particularly helpful in answering others' questions.
- **Problems or concerns with grading:** Please see **Regrade Requests** in the syllabus.
- **More sensitive personal matters:** Email the instructor directly at **djcran@iu.edu (mailto:djcran@iu.edu)** .
- **Office hours:** We will offer a mix of in-person office hours and online office hours via video conference. During online office hours, use the links in the table below to join.

Office hours will be posted soon.

Day	Time (Eastern)	Instructor	Location / Zoom
Monday	3:30 - 4:30 PM	Sahaj Singh Maini	https://iu.zoom.us/j/6802094630
Tuesday			
Wednesday	1:00 - 2:00 PM	Jihye Shin	https://iu.zoom.us/j/9302156426
	5:30 - 6:30 PM	David Crandall	3020 Luddy AI Center (beginning Jan 28)
Thursday			
Friday			

Reading assignments

The readings may refer to the following resources:

- Szeliski, *Computer Vision: Algorithms and Applications*, 2nd ed. (Springer, 2022). Available online: **http://szeliski.org/Book (http://szeliski.org/Book)**
- Burger & Burge, *Principles of Digital Image Processing: Fundamental Techniques* (2009). IU catalog: **http://www.iucat.iu.edu/catalog/12457124 (http://www.iucat.iu.edu/catalog/12457124)**
- Burger & Burge, *Principles of Digital Image Processing: Core Algorithms* (2009). IU catalog: **http://www.iucat.iu.edu/catalog/11881641 (http://www.iucat.iu.edu/catalog/11881641)**
- Burger & Burge, *Principles of Digital Image Processing: Advanced Methods* (2009). IU catalog: **http://iucat.iu.edu/catalog/13490678 (http://iucat.iu.edu/catalog/13490678)**
- Goodfellow, Bengio, & Courville, *Deep Learning* (2016). Available online: **https://www.deeplearningbook.org (https://www.deeplearningbook.org)**
- Bishop & Bishop, *Deep Learning*. Available online: **https://www.bishopbook.com/ (https://www.bishopbook.com/)** (free to download on the IU network)

Tentative schedule, readings, and resources

- Lectures are generally on marked below in red. Discussion sessions are marked below in green.
- The **Required reading** items are what you should read **before** that day's class.
- The **Optional Readings** items lists extra resources that may help you understand better, but aren't required.

This schedule is tentative and subject to change.

Mon Jan 12 Lecture: Welcome and Introduction to Computer Vision

- **Optional reading:**
  - Szeliski 1.1–1.2
- **Class session:** **Slides (https://iu.instructure.com/courses/2364992/files/200934057?wrap=1)**, **Video (https://iu.mediaspace.kaltura.com/media/t/1_6pevrxmz)**

Wed Jan 14 Discussion+Lecture: Image Filtering and Convolutions

- **Required reading:**
  - **The Affective Growth of Computer Vision (https://openaccess.thecvf.com/content/CVPR2021/papers/Su_The_Affective_Growth_of_Computer_Vision_CVPR_2021_paper.pdf)** (CVPR 2021)
- **Optional reading:**
  - Szeliski 3.2
  - Burger Fundamentals 1.2.3–1.2.6, 5.1–5.2
- **Class session:** **Slides (https://iu.instructure.com/courses/2364992/files/200934077?wrap=1)**, **Video (https://iu.mediaspace.kaltura.com/media/t/1_e54pxu32)**

Mon Jan 19: No class (University holiday)

Wed Jan 21 Lecture: Edge and Corner Detection

- **Optional reading:**
  - Burger Fundamentals Ch 6.1-6.4
  - Burger Core Ch. 3
  - Szeliski 7.2, 7.4
- **Class session:** **Slides (https://iu.instructure.com/courses/2364992/files/201165524?wrap=1)**, **Video (https://iu.mediaspace.kaltura.com/media/t/1_u7401uoo)**

Mon Jan 26 Discussion: Convolution and Convolutional Neural Networks

- **Required reading:**
  - **A guide to convolution arithmetic for deep learning (https://arxiv.org/pdf/1603.07285)**
  - **Gradient based learning applied to document recognition (http://yann.lecun.com/exdb/publis/pdf/lecun-01a.pdf)** *(but only read through section IIIB (i.e. pages 1–11 — don't read the whole thing!!))*
- **Optional reading:**
  - **Xception: Deep Learning with Depthwise Separable Convolutions (https://arxiv.org/abs/1610.02357)**
  - **Multi-Scale Context Aggregation by Dilated Convolutions**

(https://arxiv.org/abs/1511.07122)
    - Activation Functions: Comparison of Trends in Practice and Research for Deep Learning (https://arxiv.org/abs/1811.03378)
- **Optional source code:**
    - Convolutions (PyTorch) (https://github.com/vkvats/B657-ComputerVision/blob/main/Week%202%20-%20Convolutions.ipynb)
    - LeNet (PyTorch) (https://github.com/vkvats/B657-ComputerVision/blob/main/Week%203%20-%20LeNet.ipynb)
- **Class session:** Slides, (https://iu.instructure.com/courses/2364992/files/201773729?wrap=1) Video (https://iu.mediaspace.kaltura.com/media/t/1_u4ip9fl4)

Wed Jan 28 Lecture: Feature points and model fitting

- **Optional reading:**
    - Burger Core Ch 4
    - Burger Advanced Ch. 7
    - Szeliski 7.1
- **Class session:** Slides (https://iu.instructure.com/courses/2364992/files/201448348?wrap=1) , Video (https://iu.mediaspace.kaltura.com/media/t/1_iyrgj45b)

Mon Feb 2 Discussion: AlexNet and VGG

- **Required reading:**
    - ImageNet Classification with Deep Convolutional Neural Networks (https://proceedings.neurips.cc/paper_files/paper/2012/file/c399862d3b9d6b76c8436e924a68c45b-Paper.pdf)
    - Very Deep Convolutional Networks for Large-Scale Image Recognition (https://arxiv.org/abs/1409.1556)
- **Optional reading:**
    - Szeliski 5.3, 5.4
    - Batch Normalization: Accelerating Deep Network Training by Reducing Internal Covariate Shift (https://arxiv.org/abs/1502.03167)
    - Layer Normalization (https://arxiv.org/abs/1607.06450)
    - Group Normalization (https://arxiv.org/abs/1803.08494)
    - Instance Normalization: The Missing Ingredient for Fast Stylization (https://arxiv.org/abs/1607.08022)

- Micro-Batch Training with Batch-Channel Normalization and Weight Standardization (https://arxiv.org/abs/1903.10520)
- Dropout: A Simple Way to Prevent Neural Networks from Overfitting (https://jmlr.org/papers/v15/srivastava14a.html)

- **Optional source code:**
  - AlexNet (PyTorch) (https://github.com/vkvats/B657-ComputerVision/blob/main/Week%204%20-%20AlexNet.ipynb)
  - VGG (PyTorch) (https://github.com/vkvats/B657-ComputerVision/blob/main/Week%204%20-%20VGG-Net.ipynb)
  - Normalization and DropOut (PyTorch) (https://github.com/vkvats/B657-ComputerVision/blob/main/Week%204%20-%20Normalization.ipynb)
- **Class session:** Slides (https://iu.instructure.com/courses/2364992/files/201773728?wrap=1), [no video available]

Wed Feb 4 Lecture: Image transformations

- **Optional reading:**
  - Burger Core 10.1-10.3
  - Szeliski 2.1, 6.1, 8.1
- **Class session:** Slides (https://iu.instructure.com/courses/2364992/files/201727628?wrap=1), Video (https://iu.mediaspace.kaltura.com/media/t/1_mz8m17nf)

Mon Feb 9 Discussion: Inception and GoogLeNet

- **Required reading:**
  - Deeply-Supervised Nets (https://arxiv.org/abs/1409.5185)
  - Going Deeper with Convolutions (https://arxiv.org/abs/1409.4842)
- **Optional reading:**
  - Distilling the Knowledge in a Neural Network (https://arxiv.org/abs/1503.02531)
  - Deformable Convolutional Networks (https://arxiv.org/abs/1703.06211)
  - Rethinking the Inception Architecture for Computer Vision (https://arxiv.org/abs/1512.00567)
  - Network In Network (https://arxiv.org/abs/1312.4400)
- **Optional source code:**
  - Knowledge Distillation (PyTorch) (https://github.com/vkvats/B657-ComputerVision/blob/main/Week%205%20-%20Knowledge-Distillation.ipynb)

- GoogLeNet (PyTorch) (https://github.com/vkvats/B657-ComputerVision/blob/main/Week%205%20-%20GoogLeNet.ipynb)
- Network in Network (PyTorch) (https://github.com/vkvats/B657-ComputerVision/blob/main/Week%205%20-%20Network_in_Network.ipynb)

- **Class session:** Slides (https://iu.instructure.com/courses/2364992/files/201965371?wrap=1), Video (https://iu.mediaspace.kaltura.com/media/t/1_4ksornai)

Wed Feb 11 Lecture: Projections

- **Optional reading:**
  - Burger Fundamentals 1.2
  - Szeliski 2.1, 2.2.3
- **Class session:** Slides (https://iu.instructure.com/courses/2364992/files/202326173?wrap=1), Video (https://iu.mediaspace.kaltura.com/media/t/1_4elnip6l)

Mon Feb 16 Discussion: ResNet and Faster R-CNN

- **Required reading:**
  - Deep Residual Learning for Image Recognition (https://arxiv.org/abs/1512.03385)
  - Faster R-CNN: Towards Real-Time Object Detection with Region Proposal Networks (https://arxiv.org/abs/1506.01497)
- **Optional reading:**
  - Wide Residual Networks (https://arxiv.org/abs/1605.07146)
  - Deep Networks with Stochastic Depth (https://arxiv.org/abs/1603.09382)
  - Aggregated Residual Transformations for Deep Neural Networks (https://arxiv.org/abs/1611.05431) (aka ResNext)
  - Identity Mappings in Deep Residual Networks (https://arxiv.org/abs/1603.05027)
  - Rich feature hierarchies for accurate object detection and semantic segmentation (https://arxiv.org/abs/1311.2524) (aka R-CNN paper)
  - Fast R-CNN (https://arxiv.org/abs/1504.08083)
- **Optional source code:**
  - ResNet (PyTorch) (https://github.com/vkvats/B657-ComputerVision/blob/main/Week%206%20-%20ResNets.ipynb)
  - Wide ResNet (PyTorch) (https://github.com/vkvats/B657-ComputerVision/blob/main/Week%206%20-%20Wide-ResNets.ipynb)
  - ResNext (PyTorch) (https://github.com/vkvats/B657-

ComputerVision/blob/main/Week%206%20-%20ResNeXt.ipynb)
    - Stochastic ResNet (PyTorch) (https://github.com/vkvats/B657-ComputerVision/blob/main/Week%206%20-%20Stochastic-ResNets.ipynb)
- **Class session:** Slides (https://iu.instructure.com/courses/2364992/files/202326171?wrap=1), Video (https://iu.mediaspace.kaltura.com/media/t/1_ekisdix6)

Wed Feb 18 Lecture: Cameras and Color

- **Optional reading:**
    - Szeliski 2.3
    - Burger Fundamental Ch. 8
- **Class session:** Slides (https://iu.instructure.com/courses/2364992/files/202326169?wrap=1), Video (https://iu.mediaspace.kaltura.com/media/t/1_3lqlx6rc)

Mon Feb 23 Discussion: YOLO and FPNs

- **Required reading:**
    - You Only Look Once: Unified, Real-Time Object Detection (https://arxiv.org/abs/1506.02640)
    - Feature Pyramid Networks for Object Detection (https://arxiv.org/abs/1612.03144)
- **Optional reading:**
    - SSD: Single Shot MultiBox Detector (https://arxiv.org/abs/1512.02325)
    - YOLO9000: Better, Faster, Stronger (https://arxiv.org/abs/1612.08242)
    - Focal Loss for Dense Object Detection (https://arxiv.org/abs/1708.02002)
- **Class session:** Slides (https://iu.instructure.com/courses/2364992/files/202532441?wrap=1), Video (https://iu.mediaspace.kaltura.com/media/t/1_gb895649)

Wed Feb 25: Depth and Stereo

- **Optional reading:**
    - Szeliski 11.1, 11.3-11.5
- **Class session:** Slides (https://iu.instructure.com/courses/2364992/files/202769384?wrap=1), Video (https://iu.mediaspace.kaltura.com/media/t/1_75oz9g29)

Mon Mar 2 Discussion: Transformers

- **Required reading:**
  - Attention Is All You Need (https://arxiv.org/abs/1706.03762)
  - An Image is Worth 16x16 Words: Transformers for Image Recognition at Scale (https://arxiv.org/abs/2010.11929)
- **Optional reading:**
  - The Illustrated Transformer (http://jalammar.github.io/illustrated-transformer/)
  - Transformers are Graph Neural Networks (https://arxiv.org/abs/2506.22084)
- **Optional source code:**
  - Vision Transformer (PyTorch) (https://github.com/vkvats/B657-ComputerVision/blob/main/Week%207%20-%20VisionTransformer.ipynb)
  - Attention Visualization (PyTorch) (https://github.com/vkvats/B657-ComputerVision/blob/main/Week%207%20-%20Attention-Visualization.ipynb)
- **Class session:** Slides, Video (https://iu.mediaspace.kaltura.com/media/t/1_6y6j8htz)

Wed Mar 4: Stereo matching

- **Optional reading:**
  - Szeliski 12.1-12.5
- **Class session:** Slides (https://iu.instructure.com/courses/2364992/files/202970095?wrap=1), Video (https://iu.mediaspace.kaltura.com/media/t/1_wrbhaocc)

Mon Mar 9 Discussion: VITs

- **Required reading:**
  - Training data-efficient image transformers & distillation through attention (https://arxiv.org/abs/2012.12877)
  - Swin Transformer: Hierarchical Vision Transformer using Shifted Windows (https://arxiv.org/abs/2103.14030)
- **Optional reading:**
  - Swin Transformer V2: Scaling Up Capacity and Resolution (https://arxiv.org/abs/2111.09883)
  - A ConvNet for the 2020s (https://arxiv.org/abs/2201.03545) (ConvNext)
- **Optional source code:**
  - Diet Data Augmentation (PyTorch) (https://github.com/vkvats/B657-

ComputerVision/blob/main/Week%208%20-%20Diet-Data-Augmentation.ipynb)
    - Swin Transformers (PyTorch) (https://github.com/vkvats/B657-ComputerVision/blob/main/Week%208%20-%20Swin-Transformer.ipynb)
    - ConvNext (PyTorch) (https://github.com/vkvats/B657-ComputerVision/blob/main/Week%208%20-%20Conv-NeXt.ipynb)
- **Class session:** Slides (https://iu.instructure.com/courses/2364992/files/203436278?wrap=1), Video (https://iu.mediaspace.kaltura.com/media/t/1_n781qq5d)

Wed Mar 11: No class

Mon Mar 16: No class (Spring break)

Wed Mar 18: No class (Spring break)

Mon Mar 23 Discussion: Segmentation

- **Required reading:**
    - U-Net: Convolutional Networks for Biomedical Image Segmentation (https://arxiv.org/pdf/1505.04597)
    - Segment Anything (https://openaccess.thecvf.com/content/ICCV2023/papers/Kirillov_Segment_Anything_ICCV_2023_paper.pdf)
- **Optional reading:**
    - Pyramid Vision Transformer: A Versatile Backbone for Dense Prediction without Convolutions (https://arxiv.org/pdf/2102.12122)
    - Mask R-CNN (https://arxiv.org/abs/1703.06870)
    - Fully Convolutional Networks for Semantic Segmentation (https://arxiv.org/abs/1605.06211)
- **Class session:** Slides, Video (https://iu.mediaspace.kaltura.com/media/t/1_ftyqhmi8)

Wed Mar 25: Segmentation and MRFs

- **Optional readings:**

- Szeliski 4.3, 7.5
- **Class session:** Slides (https://iu.instructure.com/courses/2364992/files/203560140?wrap=1) , Video (https://iu.mediaspace.kaltura.com/media/t/1_0z92jedb)

Mon Mar 30 Discussion: Autoencoders and GANs, Poster design tips

- **Required reading:**
  - Generative Adversarial Networks (https://arxiv.org/abs/1406.2661)
  - Auto-Encoding Variational Bayes (https://arxiv.org/pdf/1312.6114)
- **Class session:** Slides, Video (https://iu.mediaspace.kaltura.com/media/t/1_n7go0jyk)

Wed Apr 1: Face detection, eigenfaces, AdaBoost, integral images

- **Optimal readings:**
  - Szeliski 5.2, 3.2.3, 6.3
- **Class session:** Slides, Video (https://iu.mediaspace.kaltura.com/media/t/1_eff7nfx0)

Mon Apr 6: Interim Project Presentations

- **Class session:** Video (https://iu.mediaspace.kaltura.com/media/t/1_p1sqlam4)

Wed Apr 8: Interim Project Presentations

- **Class session:** Video (https://iu.mediaspace.kaltura.com/media/t/1_5y5as6j4)

Mon Apr 13 Discussion: Diffusion

- **Required reading:**
  - Deep Unsupervised Learning using Nonequilibrium Thermodynamics (https://arxiv.org/abs/1503.03585)
- **Optional readings and resources:**

- ICML talk about diffusion (https://www.youtube.com/watch?v=XLzhbXeK-Os&ab_channel=AIInsightswithJason)
- Yang Song's talk about diffusion (https://www.youtube.com/watch?v=nv-WTeKRLI0)
- Yang Song's blog about diffusion (https://yang-song.net/blog/2021/score/)
- Generative Modeling by Estimating Gradients of the Data Distribution (https://arxiv.org/abs/1907.05600), and video (https://www.youtube.com/watch?v=8TcNXi3A5DI&feature=youtu.be)

- **Class session:** Slides, Video (https://iu.mediaspace.kaltura.com/media/t/1_0btqilq0)

Wed Apr 15: Recognition

- **Optional reading:**
  - Szeliski ch. 6
- **Class session:** [No video available]

Mon Apr 20 Discussion: Diffusion (cont'd)

- **Required reading:**
  - Denoising Diffusion Probabilistic Models (https://arxiv.org/pdf/2006.11239)
- **Class session:** Video (https://iu.mediaspace.kaltura.com/media/t/1_o1az149y)

Wed Apr 22: Event-based cameras

- **Class session:** Video (https://iu.mediaspace.kaltura.com/media/t/1_58heyii9)

Mon Apr 27 Discussion: 3D Gaussian Splatting

- **Required reading:**
  - 3D Gaussian Splatting for Real-Time Radiance Field Rendering (https://arxiv.org/pdf/2308.04079)
- **Class session:** [No video available]

Wed Apr 29

- **Class session:** Video (https://iu.mediaspace.kaltura.com/media/t/1_onxk4m5o)

Mon May 4: Poster session